Project Management System
Requirements Document

COSC 3380 (T/Th): Databases
Dr. Uma Ramamurthy

Team 9

Courtney Berry (1954451)

Nykolas Farhangi (1909624)

Jairo Pedroza (1892136)

Anil Shanker (1985824)

Steven Weng (1951986)

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Details of the Mini-World

Project Management System is a web work management platform designed to help companies and regular people organize, track, and manage their work.

A person creates a user account on our website. A user can either create a workspace or be invited to one. Users can be a part of multiple workspaces.

When a user joins a workspace, a workspace user is created and linked to the user. The workspace user object stores user information pertinent to the workspace, such as permission role, assigned projects, and assigned department.

In a workspace, projects and departments can be created. Projects organize tasks by theme or group. Departments can be used to organize users. Users can be given prescriptive access to certain or all projects.

Projects are organization structures that hold a title, description, and a list of tasks. Projects can have estimated budgets and estimated total effort.

A task describes a single unit of work and is the most granular unit of our system. Users create tasks and assign basic information such as a title and description and may estimate fields such as predicted effort and cost.

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Entities, Attributes, Constraints

Required attributes denoted by: ATTRIBUTE
Optional attributes denoted by: ATTRIBUTE

Task

Tasks are the basic building blocks of the Project Management System. At minimum, a task contains a title, a completion status, and metadata about user interaction.

- id: id
- project_id: project_id
- workspace_id: workspace_id
- createdAt: datetime
- createdBy: workspace_user_id
- lastUpdated: datetime
- updatedBy: workspace_user_id
- title: string
- description: string
- status: int
    - Integer status codes for each completion status
    - Statuses: In Progress, Not Started, Done, Needs Review
    - Stretch: Custom User-Defined Statuses
- effort: float
    - Estimated time spent on a task
- cost: float
    - Estimated fiscal cost of a task
- effortActual: float
- costActual: float
- startDate: datetime
- dueDate: datetime
- taskStarted: datetime
- taskEnded: datetime
- tags: string[]
- subtasks: task_id[]

Project

Tasks are organized into projects. At minimum, a project contains a title, a list of tasks, and metadata about user interaction.

- id: project_id
- workspace_id: workspace_id
- createdAt: datetime

- createdBy: workspace_user_id
- lastUpdated: datetime
- updatedBy: workspace_user_id
- title: string
- tasks: task_id[]
- customTags: string[]
- budget: float
    - Estimated total cost of the project
- effort: float
    - Estimated total effort of the project

Workspace

The top-level organization structure. Users will create or be added to a workspace. Projects, departments, and admins are owned by the workspace.

- id: workspace_id
- createdAt: datetime
- createdBy: user_id
- lastUpdated: datetime
- updatedBy: user_id
- users: workspace_user_id[]
- projects: project_id[]
- departments: department_id[]

Workspace User

Stores user information pertinent to the workspace, such as workspace role, assigned projects, and assigned department.

- id: workspace_user_id
- user_id: user_id
- workspace_id: workspace_id
- role: int
    - Specific integer codes for each role
    - Roles: Owner, Admin, Member
- projects: project_id[]
- department: department_id

Department

- id: department_id
- workspace_id: workspace_id
- createdAt: datetime
- createdBy: user_id

- lastUpdated: datetime
- updatedBy: user_id
- title: string
- description: string

User

A top-level entity. Users are independent of all other entities. A user holds personal information, preferences, and other basic information.

- id: user_id
- firstName: string
- lastName: string
- email: string
- workspaces: workspace_id[]
- phoneNumber: string

Entity Relationships

Workspace → Projects
One → Many

Workspace : Departments
One → Many

Users : Workspaces
One → Many

Workspace : Workspace Users
One → Many

Department : Workspace Users
One → Many

Project : Tasks
One → Many

Task : Tasks
One → Many

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Project Specifications

- Users create and can be added to workspaces
- In a workspace, projects and departments can be created
- In a workspace, users can be assigned to projects and a department
- In a workspace, users can be assigned one of the following roles:
    - Member: Can create, delete, and modify tasks within their assigned projects.
    - Admin: Can create, delete, and modify projects. Has access to all projects. Can add users to workspace. Can assign projects to users.
    - Owner: Can create, delete, and modify the workspace. Can assign user roles. Workspace creator by default.
- A user's assigned department has no impact on permissions
- A task's effort is estimated by the user in hours

Project Management System Conceptual Schema

